Opeyemi Peter Ojajuni, Ph.D.

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Baton Rouge, USA

EDUCATION

- **Southern University and A & M College, USA** *Aug 2018 - Dec 2023*
  Ph.D. in Science and Mathematics. | Relevant Courses: Applied Statistics, Data mining, Research Designs
- **University of Surrey, UK** *Sep 2013 – Sept 2014*
  MSc in Mobile and Satellite Communication
  Covenant University, Nigeria *Sep 2007 - July 2012*
  B.Eng in Computer Engineering

SKILLS

- **Programming skills**: Python, R, SQL, and PySpark, HTML, CSS, JavaScript.
- **Databases:** SQL, MongoDB, InfluxDB, Postgres, ChromaDB, Vector Databases, FAISS
- **Packages and Libraries:** NumPy, SciPy, Pandas, TensorFlow, Keras, Theano, Caffe, PyTorch, PyTorch, SciKit-Learn,
- **Machine Learning Framework**: Pandas, Numpy, TensorFlow, PyTorch, scikit-learn, XGBoost, LightGBM, Hugging Face, OpenCV, Tableau, Power BI, Matplotlib, Plotly
- **Artificial Intelligence and Machine Learning:** Expertise in Gen-AI and Deep Learning - LLMs, RAG, Fine-Tunning, Prompt-Tunning, Prompt Engineering, Langchain, LlamaIndex, Haystack, Multi-Agent frameworks, CrewAI, LangGraph, BERT, GPT-based models
- **Predictive Modeling and Forecasting**: Time Series Forecasting and Statistical Modeling.
- **Cloud Platforms**: AWS (SageMaker, Bedrock, Lambda, Glue, EC2), Azure ML, GCP Vertex AI

EXPERIENCE

- **Southern University and A & M College, USA** *Feb 2024 - Present*
  Postdoctoral – Applied Scientist (Data Science & Machine learning) Baton Rouge
  - Currently developing AI-driven Virtual Reality (VR) systems that integrate Large Language Models (LLMs) and retrieval-augmented generation (RAG) pipelines using well-known frameworks like Langchain and LlamaIndex with cognitive modeling, LiDAR/GIS-based digital twins, and immersive hazard simulations to enhance real-time emergency response, situational awareness, and decision support.
  - Led applied research initiative blending VR software application development, UX design, and educational data analytics to bridge the digital divide; reached 500+ students and 100+ STEM educators, increasing STEM engagement (+47%), concept retention (+32%), and teacher technology self-efficacy (+41%) through multi-user immersive learning modules.
  - Designed, trained, and benchmarked supervised ML models (SVM, Random Forest, XGBoost, KNN) to predict Computational Thinking proficiency using survey and performance datasets; achieved 92% predictive accuracy, identified 5 latent cognitive factors via Principal Component Analysis (PCA), and Structural Equation Modeling (SEM), and reduced assessment misclassification by 21%.
  - Developed InsightBot, a GenAI-powered research synthesis platform integrating LangChain, OpenAIEmbeddings, and FAISS for retrieval-augmented generation (RAG); processed multi-article queries with 92% retrieval precision and 1.8-second average response latency, reducing manual review time by 80%.
  - Mentored and supervised 10+ junior researchers in responsible AI/ML application design, RAG architecture, and XR system evaluation, increasing lab publication output and model performance reproducibility by 50%.

- **Amazon.com** *June 2023 - Sept 2023*
  Data Scientist Intern Seattle, WA
  - Performed large-scale temporal analysis on 1B+ historical supply chain transactions and customer behavior using advanced statistical and ML models, improving packaging risk detection and simulation robustness by 40%, directly supporting sustainable logistics at global scale.
  - Applied clustering and behavioral segmentation to 10M+ Amazon customer orders, optimizing packing capacity planning and reducing fulfillment bottlenecks by 23%.
  - Developed a data-driven simulation approach using probability density functions and empirical distribution fitting to generate realistic datasets boosting packing time forecast accuracy by 40% and enabling more reliable cost variability analysis for strategic risk management.
  - Collaborated across engineering, operations, and product teams to align risk modeling with sustainability objectives, translating complex analytical insights into actionable business recommendations for global impact.

- **Hacey initiatives** *Jan 2018 - Present*
  Consultant- Data Scientist / Machine Learning Engineer Remote
  - **Projects**: Consultancy in application of AI for Financial Assistant System
  - **Data Engineering:** Processed and structured 10 years of financial literacy data (~2,500 pages) using PyMuPDF and Pandas, transforming it into a semantically indexed knowledge base with LangChain pipelines, Titan embeddings, and FAISS for fast and accurate retrieval.

- **RAG-powered AI System:** Built and deployed a Streamlit proof-of-concept application that delivered real-time question answering and explainable natural language responses, demonstrating the value of personalized financial guidance for high-risk user segments.

- **Southern University and A & M College** *Aug 2018 - Dec 2023*
  Graduate Research Assistant Baton Rouge, LA
  - Conducted a controlled experimental study using validated CT instruments and statistical analyses to evaluate pre/post VR learning effects, revealing 31% overall gains in cognitive skills
  - Conducted controlled VR user experience research studies across 60+ participants, combining UX research, usability testing, and presence measurement to quantify cognitive load and user immersion, leading to a 35% reduction in task completion errors in the final VR prototype.
  - Built and deployed an academic success prediction platform using supervised ML models (Random Forest, XGBoost, SVM) with 92% predictive accuracy, integrating MLOps pipelines for real-time cognitive risk detection and adaptive feedback in VR learning systems.
  - Modeled education-to-workforce transition patterns using ARIMA, producing forecasts that informed strategic recruitment policies in emerging STEM fields.

PROJECTS

- **Real Estate ML/DL interactive Web App**: Predict house prices by building ML/DL models; Model deployment using StreamLit and API built in Python.
- **EDA of Big Data using PySpark**: Open-source python library with Apache Spark for Big Data Validation using Regex in Python. Extensive Data cleaning and validation of data from multiple sources.
- **Student Academic Success Prediction ML/DL interactive Web App**: Predict student success by building ML/DL models; Model deployment using StreamLit and API built in Python.
- **Research RAG App using Python, Streamlit, LangChain, OpenAI, and FAISS:** extracts articles from user-provided URLs, splits and embeds the text into a FAISS vector database and enables retrieval-augmented question answering with source citations through an interactive web interface.

  GitHub link to Projects, **OR** View Apps on my Portfolio, please click

CERTIFICATIONS

- Amazon Web Service (AWS) Certified AI Practitioner [LINK] July 2025
- IBM Artificial Intelligence Practitioner [LINK] Jan 2022